

In Lab Tasks

You are working as a **Network Administrator** for a small office. The office has **4 computers** that need to communicate with each other.

Your manager asks you to create the network in **Cisco Packet Tracer**.

Requirements:

- 4 PCs
- 1 switch
- All PCs must communicate with each other.
- Use appropriate Ethernet connections.
- Assign suitable IP addresses.

Task:

Create the topology, configure the IP addresses, and verify connectivity between all PCs using the ping command

You're asked to compare **Hub vs Switch** behavior using Simulation Mode.

Connect 2 PC with a Switch with the same network IP Address

Connect 2 pc with a Hub with the same network IP Address.

You send one ping between two PCs behind a Hub and observe the ICMP packet appears on **every** port's event line, while the same test behind a Switch shows the ICMP packet going **only** to the destination port.

As the person explaining this to a junior colleague, what does this reveal about how hubs and switches operate at Layer 1 vs Layer 2?

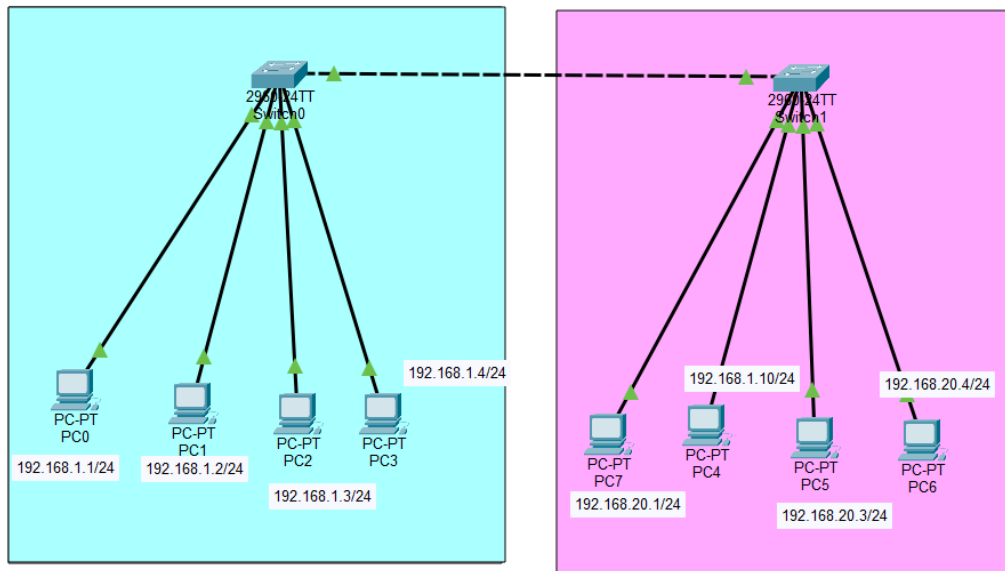
Design and configure the network in the figure below.

Check the connectivity using the SIMPLE PDU or Ping. (Do it for two different PC's)

Both for real-time and Simulation mode.

Show the ARP packet header format in Cisco Packet Tracer. (Two PC's)

Note: Show each step in the form of screenshots.



You are a **Network Administrator** for a company with 20 workstations. Manually assigning IP addresses to every workstation would be time-consuming. Your manager asks you to deploy a **DHCP server** in Packet Tracer.

Requirements:

- 1 Server
- 1 Switch
- 10s PCs
- Configure the server as DHCP server.
- PCs should obtain their IP addresses automatically.

Task:

1. Assign an IP address to the server.
2. Configure the DHCP service.
3. Create an appropriate DHCP pool.
4. Configure all PCs to use DHCP.
5. Verify that each PC receives an IP address automatically.

The lab describes DHCP as distributing IP addresses from a server pool and demonstrates configuring PCs to obtain addresses through DHCP
